Kubernetes Cluster.

1. Control plane (Master node)
2. Worker Nodes

Master Node	Worknode.
1. Kube-API / API Server 2. Kube Schedule 3. Kube Controller manager 4. Kube-Etcd	1. pod 2. Container Runtime 3. Kubelet 4. Kube-proxy.

- master node controlls entire Kubernetes cluster
- Work node receives order From master node & coordinates within master node.

Kube API - master node's central management Component of Kubernetes.
(Frontend)

Kube API → request → API → Kube Shedule

Kube Scheduler will assign the request from user Based on the nodes (Worker nodes) ~~Act~~ health check & availability

Kube Controller
1. Replicaset Controller [one Container fail aanalum adha delete panni relaunch Pannum]
2. deployment "
3. Job " - Batch management / maintain
4. node " - have both working load if in case one is not working

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Kube ETCd - all metadata are assigned for it

→ Components

- pod. Container inside the pod Shares volume, network
- Container Runtime.
- Kublet
- Kube proxy - External traffic handling.

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→ Why networking is Important in Kubernetes

⇒ Containers are dynamic and Short-lived

⇒ pods can be Created and destroyed anytime

⇒ Applications must communicates reliably

⇒ Kubernetes provides a cluster-wide networking model

→ Kubernetes Networking model

- Each pods gets a Chique IP address
- pods Communicates directly without NAT
- Nodes can Communicates with pods
- Containers in the same pods share the same Network namespace

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⇒ What is CNI [Container network Interface]

- CNI is a standard used to configure networking for containers
- It assign IP address to pods
- It connects pods to the cluster network
- Example : calico , Flannel , cilium.

⇒ What is a Kubernetes service ?

- It provides stable network endpoint for pods
- pods are temporary and their IPs change
- service provide a fixed IP and DNS name
- load balance across multiple pods.

⇒ Types of Kubernetes service

1. Cluster IP - Internal Communication within Cluster
2. Nodeport - Exposes service on node IP and port
3. Loadbalancer - cloud provider external load Balancer
4. External name - maps service to External DNS

⇒ Role of kube-proxy.

1. Runs on every node.
2. Watches service and endpoints
3. creates iptables or IPVs rules
4. Implements service virtual IP routing

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Kubectl.

1. ~# Kubectl version [check client & server version]

2. ~# Kubectl get nodes

3. ~# Kubectl describe node k8n = master

4. ~# Kubectl run set pod -- image = httpd
   Podname

5. ~# ~~Kubernet~~
   ~# Kubectl delete deployment <name1> <name2>

6. Kubectl exec -it <podname>

7. Kubectl logs <pods_name>

8. Kubectl delete pod <pod name>

9. Kubectl create deployment <pod name> -- image = httpd = latest
   → scale var

10. Kubectl scale deployment <podname> -- replicas = 5

11. Kubectl create deployment <test> -- image = nginx -- replicas = 5

    Kubectl -- replicas = 2

    Kubectl describe pod <podname>

    → Kubectl set image deployement <test deploy> httpd = httpd trixx

    Kubectl rollout undo deployement <test deploy>

    Kubectl rollout status deployment <test deploy>

    Kubectl expose deployment <test deploy -- type = = Nodeport -- port = = 80

    Kubectl get svc

deleting deployment

deploy pod

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Understanding Kubernetes Objects

- Kubernetes object are presistent entities in the Kubernetes uses these entities to represent the state of you cluster. specially. they can describe

⇒ What containerized application are running

⇒ The resource available to those application

⇒ The policies around how those application behave such as restart policies, upgrade and fault-tolerance

deployment.yaml

```
apiversion : apps/v1
kind : Deployment
metadata :
    name : ngix-deployment
spec :
    selector :
        matchLabels :
            app: nginx
    replicas : 2
    template :
        metadata
            labels :
                app : nginx
        spec :
            Containers :
                - name : nginx
                  image : nginx : 1.14.2
                  Ports :
                    - Containerport : 80
```

Pod.yaml

```
apiversion : v1
kind : pod
metadata :
    name : ngix
spec :
    Containers
        -name : ngix
         image : ngix : 1.14.2
         Ports :
          - Container Port : 80
```

Cmt

→ Kubectl apply -f pod.yaml

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⇒ Defining a service.

apiversion : v1

Kind : service

metadata :

name : my - service

spec :

selector :

app. kubernetes. io / name : MyApp

Ports :

Protocal : TCP

Port : 80

target port : 9376

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